

Course III: Corporate Finance

Summer Term 2026

Unit 2 Mini-Cases

Please upload your solutions and indicate the Mini-Cases you have solved before 10:00 p.m. on March 22, 2026. There will be no deadline extensions!

To upload your solutions and make indications, please go to the assignment section on Canvas and choose “Unit 2 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). Only complete submissions containing answers to the T/F questions and a solutions upload will be considered. Your last complete attempt counts.

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solution file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Mini Case 2.1: Consider an unlevered firm with a value of

$$V_U = \text{EUR 200 million}$$

The firm can issue permanent debt. The possible debt levels and the expected present value of financial distress costs at those levels are:

- $D = 0$ with associated distress cost 0.
- $D = 50$ with associated distress cost 1.
- $D = 100$ with associated distress cost 5.
- $D = 150$ with associated distress cost 14.

Tax rates:

- corporate tax rate $T_C = 30\%$
- personal tax rate on interest income $T_D = 37.5\%$
- personal tax rate on equity payouts $T_E = 15\%$

Questions:

- (i) Compute the effective tax advantage of debt T_g .
- (ii) Compute the levered firm value at each debt level.
- (iii) Which debt level maximizes firm value?
- (iv) Repeat part (ii) assuming no personal taxes. How does the optimal debt level change?

Mini Case 2.2: DRT Industries has just issued EUR 50 million in perpetual debt at par (i.e., the current interest rate equal bond's coupon rate) with no maturity date. The firm will pay interest only on this debt. DRT's marginal tax rate is 15% for the foreseeable future.

- (i) Suppose the current interest rate on this debt is 2% per year. What is the annual interest tax shield and the present value of the tax shield, assuming its risk is the same as the debt?

Ten years have passed since the issuance. DRT's tax rate has remained the same, but interest rates have dropped to 1.5%, so the cost of debt capital is 1.5%.

- (ii) What is the new annual interest tax shield and its present value?
- (iii) An alternative interpretation of a "perpetual bond" is that a bond is issued with certain maturity, but the firm plans to refinance maturing debt with new debt forever. What would the annual interest tax shield and its present value be in this case?

Mini Case 2.3: A company has identified a new investment opportunity. Depending on the success of the new investment, the company will have a value of \$200 million with probability of 40% or \$105 million with probability 60% next year. You have already evaluated that the company's value without leverage would be \$130 million.

The company is considering financing the investment cost of \$105 million with a zero-coupon bond. As debtholders require a return of 6%, the company issues a zero-coupon bond with a face value of \$120.75 million.

Assume perfect capital markets without taxes.

- (i) What is the company's total value with leverage?
- (ii) What is the initial value of the company's equity? What is the cost of equity?

Now suppose that in the event of default 20% of the value of the assets will be lost to bankruptcy costs. The default can occur for some company-specific reasons (idiosyncratic risk), so the investors do not require an additional risk premium.

- (iii) Calculate the face value of debt that the debt holders require in order to lend \$105 million.
- (iv) What is the initial value of the company's equity?
- (v) What is the company's total value with leverage? How much value is lost through (expected) bankruptcy costs? Who bears these costs?

Mini Case 2.4: Consider an economy with aggregate corporate pre-tax cash flow of 300 per period in perpetuity. The risk-free rate is 10% and the corporate tax rate is 50%. There are three investor groups that differ only in their personal tax rate on interest income: Poor (0%), Middle-class (40%), Rich (60%). Assume no personal taxes on equity payouts for this question unless implied below.

- (i) If all corporations are completely equity-financed,
 - a. What is the total annual tax bill?
 - b. What is the combined value of all firms?
 - c. How much is equity worth in total?
- (ii) Suppose firms issue a small amount of risk-free perpetual debt such that aggregate annual interest expense rises by EUR 1 (so taxable income falls by EUR 1).
 - a. Which investor group would hold this debt in equilibrium and why?
 - b. What interest rate would the debt pay?
 - c. By how much does aggregate firm value change? (Compute the present value of the resulting tax shield.)

- (iii) Now let firms choose a constant outstanding stock of risk-free perpetual debt with market value D (economy-wide). Let E denote the total equity value. In equilibrium, one investor group holds all the debt and the other two hold equity, so funds invested satisfy $E = 2D$.
 - a. Which group holds the debt and why?
 - b. Write and solve for E and D using the identity above.
 - c. Compute total firm value and annual corporate tax revenue.
- (iv) Could a second investor group be induced to hold debt? If so, for each of the remaining groups, find the gross interest rate r on debt that makes them indifferent between holding debt and equity. Comment on whether it's plausible that all three groups hold debt in equilibrium once you consider the cost of capital versus the tax advantage.

Mini Case 2.5: A firm will pay out next period either a bad-state total cash flow of EUR 50 or a good-state total cash flow of EUR 150, each with probabilities 30% and 70% respectively. The risk-free rate is 10% and investors are risk-neutral.

- (i) Consider two financing choices:
 - All-equity finance.
 - Debt finance: issue priority debt that pays bondholders EUR 50 in the bad state and EUR 60 in the good state; shareholders receive the residual.
 - a. Create a payoff table to share- and bondholders under each financing choice.
 - b. Compute the market value of debt, equity, and the firm under each choice.
- (ii) Risk adjustment with CAPM.

Now assume investors are risk-averse. A comparable firm has equity beta of 1.2, the market return is 15%, and the risk-free rate remains 10%. Using a CAPM risk-adjusted discount rate, recompute the value of the firm from part (i). Briefly explain the change relative to the 10% discounting in (i).

- (iii) Corporate and personal taxes; perpetuity of cash flows.

Suppose from next period onward the firm's pre-tax cash flow is EUR 100 per period forever with certainty. The risk-free rate is still 10%. Corporate tax is $T_C = 20\%$, the personal tax rate on interest income is $T_D = 30\%$, and the personal tax rate on equity payouts (dividends) is $T_E = 25\%$. Consider a perpetual outstanding debt of EUR 60 paying a 10% coupon every period (i.e., EUR 6 each period).

For each financing strategy in (i):

- a. Compute the after-tax cash flow accruing to investors.
- b. Determine the appropriate discount rate given the personal taxation of interest.
- c. Compute the value of equity and the total firm value.
- d. Compute the tax shield from using debt.

Mini Case 2.6: Value company A using company B and company C as comparison firms. Investors believe that company A's risk is similar to an equally weighted combination of B and C.

Assume a debt beta is 0 and ignore taxes when unlevering betas. Company A generates an expected after-tax free cash flow of EUR 60 million at $t = 1$ with no further cash flows from existing assets.

- Company B: equity beta of 1.0 and leverage ratio of 0.5.
- Company C: equity beta 1.6 and leverage ratio of 1.0.

The risk-free rate is 4%. The expected market return is 10%. The corporate tax rate is 25% and company A plans to maintain a constant debt to capital ratio of 40%. The cost of debt is 5%.

In addition, there is a 20% probability that financial distress occurs for A at $t=1$. If distress occurs, firm value is reduced by EUR 8 million at $t=1$.

- (i) Compute the unlevered asset beta of company B and company C.
- (ii) Compute the unlevered asset beta of company A.
- (iii) Compute the unlevered cost of capital of company A.
- (iv) Compute the unlevered value of company A.
- (v) Compute the levered value of company A using the APV method.
- (vi) Compute the after-tax WACC of company A and verify the same levered value.
- (vii) Compute the net levered value of company A after expected distress costs.